

SEDIMENT: Poisoning-Resistant Behavioral Baselines for Host Intrusion Detection

Vurmusum*

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Abstract

Behavioral host intrusion detection systems learn a per-process baseline from a window of observations and score live behavior as drift against the baseline. Every such system rests on an implicit assumption that the learning window is free of adversary activity. In practice attacker dwell time is nonzero by definition, and a detector that turns on while the adversary is already on the host fits a baseline shaped by the adversary. We call this *baseline poisoning*. We formalize the threat model with an adversary budget $\beta \in [0, 1]$ that bounds the fraction of the window the adversary controls, propose SEDIMENT, a per-feature trimmed-moment estimator with a closed-form Gaussian-truncation bias correction and a single trim-fraction knob τ that gives defensible resistance up to $\beta = \tau/2$, and evaluate across nine experiments. Under uniform poisoning at $\beta = 0.15$ on an isotropic synthetic workload, the naive mean-and-stddev estimator loses roughly a factor of 3.6 in target-score discriminability while SEDIMENT at $\tau = 0.30$ holds. Under a grid-search white-box adversary with full knowledge of τ , the optimal placement reduces to the at-target injection, which gives evidence that white-box knowledge does not yield a stronger attack against symmetric-trim defenders. On a real captured trace of Linux `bash` activity, SEDIMENT detects an attack-payload run with a $126\times$ median score separation over clean windows and zero false positives during the attack capture period. A reference Python implementation and a production Rust runtime stay bit-equivalent through cross-implementation parity tests on every push.

1 Introduction

Behavioral host intrusion detection systems watch a process’s syscalls, file access, network endpoints, and child spawns, fit a per-process baseline from those observations, and score live behavior as drift against the baseline. The lineage starts with Forrest’s “sense of self” [1] and runs through n-gram sequence models, hidden-Markov detectors, and modern provenance-graph systems [2]. The features changed; the assumption that the learning window is adversary-free did not.

That assumption fails in practice. Mandiant’s M-Trends 2025 report [3] puts the global median attacker dwell time at 11 days for 2024 incidents, with a longer tail for sophisticated threat actors. Any detector that turns on while the adversary is already on the host fits a baseline shaped by the adversary. After that, the adversary’s behavior *is* the baseline’s notion of normal, and the detector cannot tell planted behavior apart from genuine drift.

We call this *baseline poisoning*. The lever is not exotic. A single elevated shell session can shape the baseline of every behavioral feature the detector tracks for that process, and a carefully-paced injection slips past any change-point detector watching the live stream. The question is what property an estimator needs to stay useful when the learning window is partially adversarial, and what an operator pays for that property in clean-condition false-positive rate.

This paper makes four contributions.

1. We **formalize** the baseline-poisoning threat model (section 3). The adversary has a budget $\beta \in [0, 1]$ that bounds the fraction of the learning window they control. We consider observability variants up to a white-box adversary that knows the defender’s hyperparameters.

*Independent. Email: vurmusum@gmail.com.

2. We **demonstrate** that the naive mean-and-stddev estimator widely used in prior work loses roughly a factor of 3.6 in target-score discriminability at a 15% poisoning budget on our isotropic synthetic workload, and continues to degrade smoothly as the budget grows (section 6.3).
3. We **propose** SEDIMENT (section 5), a per-feature trimmed-moment estimator with a closed-form Gaussian-truncation bias correction. Setting $\tau \geq 2\beta$ gives a defensible robustness guarantee against any adversary with budget at most β .
4. We **evaluate** the design space across nine experiments (section 6), including a four-way ablation against alternative robustness primitives and three poisoning attacks: uniform shuffled, contiguous burst, and a grid-search white-box adversary that knows τ .

A reference Python implementation accompanies a production Rust runtime; cross-implementation parity tests on every push keep them bit-equivalent across both the estimator and the syscall n-gram extractor. The artifact is open source.¹

2 Background and related work

2.1 Behavioral host intrusion detection

Forrest’s “sense of self” [1] observed that a process’s syscall sequence under normal operation is a stable, low-entropy fingerprint and that deviations from that fingerprint are detectable. Subsequent work moved along two axes. *Richer features*: from raw syscall sequences to file-access profiles, network endpoints, and child-spawn graphs. *Richer models*: from fixed-length n-grams to hidden-Markov models, recurrent networks, and provenance-graph anomaly detectors [2]. The modeling choices have moved a lot in thirty years. The assumption that the learning window is adversary-free has not.

2.2 Mimicry attacks

A long line of work shows that detectors which inspect only the live behavioral stream fall to mimicry attacks: an adversary who knows the detector’s signature pads malicious activity with otherwise benign behavior to evade detection [4, 2]. Goyal et al. force misclassification on every provenance-graph host intrusion detector they test by embedding substructures from legitimate process activity into the attack subgraph [2].

Baseline poisoning is related but distinct. Mimicry adversaries shape what their attack *looks like* at detection time. Poisoning adversaries shape what the detector considers *normal* at training time. The two compose: an adversary who can do both has more leverage than one who can only do one.

2.3 Robust statistics

Robust statistics gives several primitives for estimating central tendency and dispersion in the presence of outliers [5, 6]. SEDIMENT draws on the trimmed-mean tradition. The classical breakdown point of the symmetric τ -trimmed mean is $\tau/2$: an adversary controlling fewer than $\tau N/2$ samples cannot move the trimmed mean an arbitrary amount [5]. The piece we add is a closed-form Gaussian-truncation correction that recovers an unbiased estimate of the underlying σ on clean data while keeping the trimmed mean’s resistance.

2.4 Concept drift versus adversarial drift

Production behavioral detectors often conflate two distinct things under the label “drift”: natural *concept drift* (the workload genuinely changed) and *adversarial drift* (an attacker is manipulating the inputs). CHRONOSYND treats them as orthogonal. The maintenance-window mechanism in section 4 addresses concept drift by giving operators an explicit way to mark intervals where baseline shifts are expected. SEDIMENT addresses adversarial drift at the estimator level.

¹<https://github.com/ChronosynD/core>

3 Threat model

Setting. A behavioral HIDS observes a per-process feature stream and fits a baseline from N benign-window observations $X = \{x_1, \dots, x_N\} \subset \mathbb{R}^d$. After fitting, it scores live observations against the fitted baseline and raises an alert when the score exceeds a threshold θ .

Adversary capability. The adversary controls a fraction $\beta \in [0, 1]$ of the learning window. They may inject up to $|A| = \lfloor \beta N \rfloor$ observations of their choosing, giving the defender $X' = (X \setminus B) \cup A$ where B is some discarded subset of the original benign observations.

Adversary goal. Given a target attack behavior $m^* \in \mathbb{R}^d$ representing the post-learning malicious action the adversary intends to perform, the adversary picks A to minimize the score $s(m^*)$ assigned by the baseline fitted on X' . The adversary wins if $s(m^*) < \theta$.

Observability. We consider three variants:

Black-box The adversary knows there is a baseline-fitting step but not the algorithm or hyperparameters. Realized by `inject_pre_seed`.

Grey-box The adversary knows the algorithm family but not the exact hyperparameters. Treated as a special case of white-box for evaluation.

White-box The adversary knows the algorithm and all hyperparameters, including τ . Realized by `inject_targeted`, which grid-searches the per-feature injection placement that minimizes $s(m^*)$ after a fresh fit.

Out of scope. Kernel-level compromise that lets the adversary forge process identity (PID, UID, cgroup) or backdoor the BPF runtime is out of scope. The threat model assumes the kernel-side identity surface is trustworthy. Post-fit tampering with the persisted baseline store is a separate concern, addressed in the runtime by a SHA-256 hash chain over an append-only audit log (section 4). It is orthogonal to the algorithmic contribution of this paper.

4 The CHRONOSYND system

4.1 Pipeline

CHRONOSYND is a one-directional pipeline:

kernel BPF $\longrightarrow$ ring buffer $\longrightarrow$ collector $\longrightarrow$ features $\longrightarrow$ scorer $\longrightarrow$ alerts

Kernel-side BPF programs emit raw syscall events into a userspace ring buffer. The collector normalizes them into typed events and feeds disjoint per-process windows to the n-gram feature extractor. When a window closes, the extractor emits a fixed-shape feature vector. The scoring engine compares it against the cached baseline for that process and raises an alert when the drift score exceeds the per-process threshold. A SQLite store persists baselines and is read at startup to warm the scorer’s in-memory cache.

4.2 Cross-language reference and production split

SEDIMENT and the n-gram extractor each have two implementations: a Python reference in `chronosynd-py/` and a Rust production port in `chronosynd-rs/`. Python is authoritative during research iteration. Rust is authoritative for deployment. CI runs cross-implementation parity tests on every push, covering both the baseline estimator (15 cases at three sample sizes and four estimator configurations, 20 score inputs each) and the syscall n-gram extractor (4 event-stream cases at varied window sizes and vocabularies). Divergence between sides fails the build.

4.3 Tamper-evident store

Every state-changing operation (`put_baseline`, `start_maintenance_window`, `end_maintenance_window`) commits in a single transaction together with an audit row. The audit row’s SHA-256 hash covers the operation, the payload, and the previous row’s hash. The chain is anchored at the all-zero hash, so any tampering breaks the chain at the tampered row. `verify-store` walks the chain and exits with a distinguished code on tamper detection, so an operator’s monitoring system can tell tampering apart from a CLI crash.

4.4 Operator surface

Operators interact with CHRONOSYND through a single CLI binary. The subcommands cover baseline inspection, tagging maintenance windows when planned changes will shift the baseline, walking the audit log to detect tampering, fitting baselines from CSV observations or recorded JSONL event traces, and running the collector daemon end-to-end against a configured store.

5 SEDIMENT

5.1 Algorithm

SEDIMENT models each feature as an independent Gaussian under the clean null. The score on a new observation x is the sum of squared standardized residuals,

$$s(x) = \sum_{j=1}^d \frac{(x_j - \mu_j)^2}{(\sigma_j + \epsilon)^2}, \quad (1)$$

which is the Mahalanobis distance under a diagonal-covariance assumption. The estimator has two parameters: a trim fraction $\tau \in [0, 1)$ and a small $\epsilon > 0$ that floors the variance.

For each feature j , given a learning window $\{x_1^{(j)}, \dots, x_N^{(j)}\}$ sorted ascending, the per-tail trim count is $k = \lfloor \tau N/2 + 1/2 \rfloor$. The trimmed mean is the mean of the surviving $N - 2k$ samples,

$$\mu_j = \frac{1}{N - 2k} \sum_{i=k+1}^{N-k} x_{(i)}^{(j)}. \quad (2)$$

The trimmed standard deviation is the unbiased sample standard deviation of the surviving samples, multiplied by a closed-form Gaussian-truncation correction $c(\tau)$ derived below,

$$\sigma_j = c(\tau) \cdot \sqrt{\frac{1}{N - 2k - 1} \sum_{i=k+1}^{N-k} (x_{(i)}^{(j)} - \mu_j)^2}. \quad (3)$$

5.2 Gaussian-truncation correction

For an $\mathcal{N}(0, s^2)$ sample symmetrically trimmed at $\tau/2$ per tail, the trimmed-sample variance underestimates s^2 because the trimmed tails carried disproportionate variance. Let Φ^{-1} and ϕ denote the standard normal quantile and density functions. The trim cutoff is

$$z = \Phi^{-1}\left(1 - \frac{\tau}{2}\right). \quad (4)$$

The variance of an $\mathcal{N}(0, 1)$ variable conditioned on $|Z| \leq z$ is

$$\text{Var}(Z \mid |Z| \leq z) = 1 - \frac{2z\phi(z)}{1 - \tau}, \quad (5)$$

so the multiplicative correction that recovers an unbiased estimate of s from the trimmed sample standard deviation is

$$c(\tau) = \frac{1}{\sqrt{1 - 2z\phi(z)/(1 - \tau)}}. \quad (6)$$

5.3 Robustness guarantee

The classical breakdown point of the symmetric τ -trimmed mean is $\tau/2$ [5]: an adversary controlling fewer than $\tau N/2$ samples per feature cannot move the trimmed mean an arbitrary amount. We adopt this as our defender’s contract.

Proposition 1 (SEDIMENT robustness, informal). *Fix a per-feature trim fraction τ and an adversary budget $\beta \leq \tau/2$. Let $\mu_j^\circ, \sigma_j^\circ$ denote the population mean and standard deviation of feature j under the clean data-generating distribution, and let $\hat{\mu}_j, \hat{\sigma}_j$ denote the moments fitted by SEDIMENT on a poisoned learning window of size N . Then for each feature j , $\hat{\mu}_j \rightarrow \mu_j^\circ$ and $\hat{\sigma}_j \rightarrow \sigma_j^\circ$ in probability as $N \rightarrow \infty$, regardless of the adversary’s choice of poisoning placement.*

The argument has two ingredients. First, the trimmed mean’s breakdown point of $\tau/2$ [5] bounds the influence of any βN adversarial samples on $\hat{\mu}_j$ when $\beta \leq \tau/2$, because the adversarial samples cannot all sit in the surviving region simultaneously. Second, the closed-form correction $c(\tau)$ restores the trimmed standard deviation to an unbiased estimate of σ_j° on the surviving $(1 - \tau)N$ benign samples.

The bound is tight in the worst case. At $\beta = \tau/2$, an adversary can in principle place every adversarial sample at the defender’s trim boundary, surviving the trim and dragging $\hat{\mu}_j$. section 6.10 reports a grid-search white-box adversary that attempts exactly this. Empirically the optimum reduces to the at-target injection, which gives evidence that the worst case is hard to construct in practice. We return to residual risks in section 7.

5.4 Complexity

Per feature, fitting requires sorting the N samples ($\mathcal{O}(N \log N)$) and computing the trimmed moments ($\mathcal{O}(N)$). Memory is $\mathcal{O}(d)$ for the fitted moments per process. Scoring is $\mathcal{O}(d)$ per observation.

6 Evaluation

6.1 Setup

Workloads. Three synthetic factories drive the experiments: `isotropic_gaussian` for the canonical experiments, `heterogeneous_gaussian` for per-feature heterogeneity, and `gaussian_mixture` for multimodal benign distributions. All are deterministic given a seed.

Attacks. Three poisoning attacks instantiate the threat model:

- `inject_pre_seed`, the black-box adversary, target-valued injections shuffled uniformly across the window.
- `inject_burst`, contiguous-block injections at a chosen or randomly-drawn position.
- `inject_targeted`, the white-box adversary that grid-searches the per-feature placement minimizing $s(m^*)$ after a fresh fit by an estimator with known τ .

Estimators. Four estimators are compared. NAIVE uses per-feature mean and standard deviation, the prior-work reference. CONSENSUS splits the learning window into four sub-windows and drops features whose sub-window means disagree by more than three pooled standard deviations. ANOMALY-WITHIN fits naive on the full window, drops the top decile of within-window outliers, and refits on what survives. SEDIMENT runs at trim fractions $\{0.1, 0.2, 0.3, 0.5\}$.

Reporting. Every cell uses ten independent seeds, drawn deterministically from `numpy.random.SeedSequence(seed).spawn` to give independent learning-window, poisoning, benign-test, and malicious-test streams. The target FPR is 0.05 throughout. Error bars are standard error across seeds. Every figure regenerates from `scripts/reproduce_all.sh`.

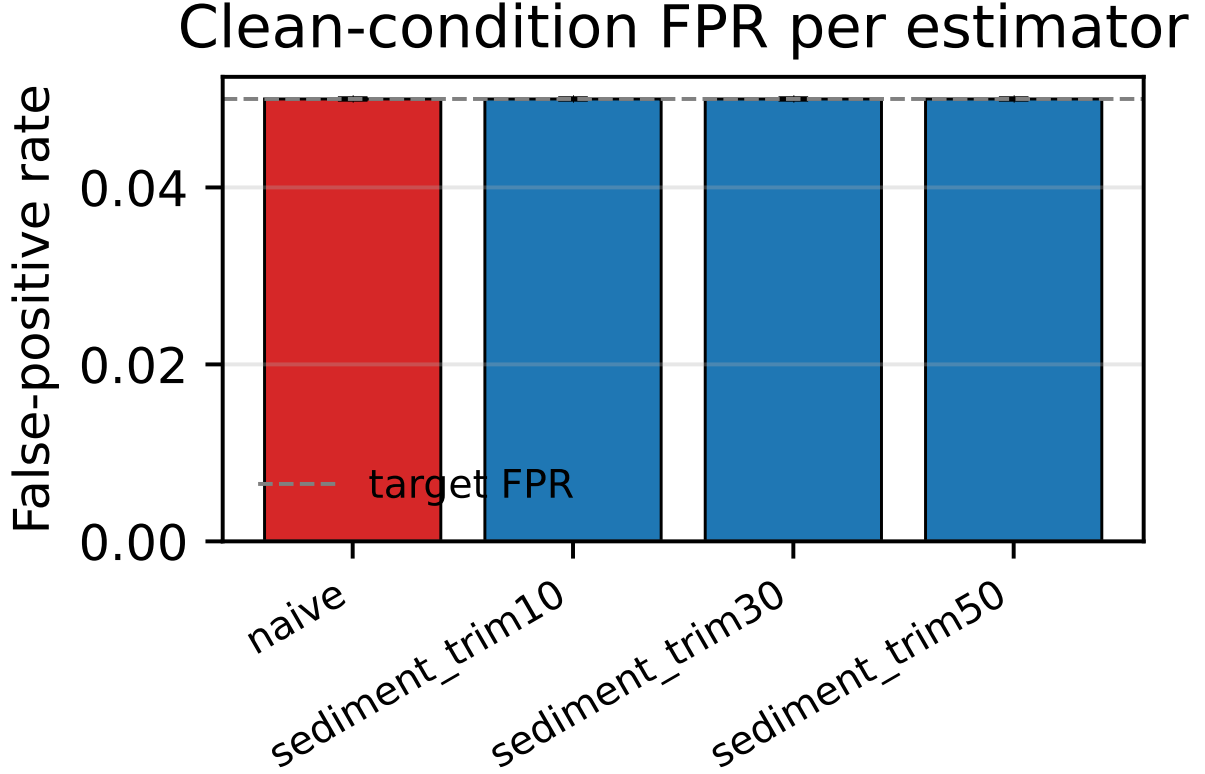


Figure 1: Clean-condition false-positive rate per estimator. Error bars are standard error across ten seeds.

6.2 Clean-baseline FPR is at the configured target

By construction the per-estimator threshold is the $1 - \text{target_fpr}$ quantile of benign scores on each cell, so the empirical FPR matches the configured target up to discretization noise. fig. 1 confirms this: every cell lands at $\text{FPR} = 0.0500$ with standard error below 0.0001 across ten seeds. This is a sanity panel; differentiation across estimators shows up in the next subsections.

6.3 Naive collapse under poisoning

fig. 2 shows the target score NAIVE assigns to m^* as the adversary’s budget grows. Clean-condition score is roughly 130. At $\beta = 0.05$ it falls to 70; at $\beta = 0.10$ to 48; at $\beta = 0.15$ to 36. The collapse is smooth and approximately log-linear in β , not threshold-like. There is no “safe” small budget. Even modest poisoning rates the adversary could plausibly achieve cost substantial discriminability.

6.4 SEDIMENT design space

fig. 3 sweeps the joint (τ, β) grid for SEDIMENT. Two structural features stand out. First, the robustness boundary tracks $\beta = \tau/2$ predicted by proposition 1: for budgets at or below half the trim fraction, the target score holds within a small constant factor of the clean-condition value. Second, the cost of choosing too high a trim is bounded: a $\tau = 0.50$ defender pays nothing in clean-condition discriminability (section 6.6) but loses headroom against budgets above 0.25. The recommended operating point is the smallest τ at least double the worst expected adversary budget.

Naive baseline collapse under poisoning

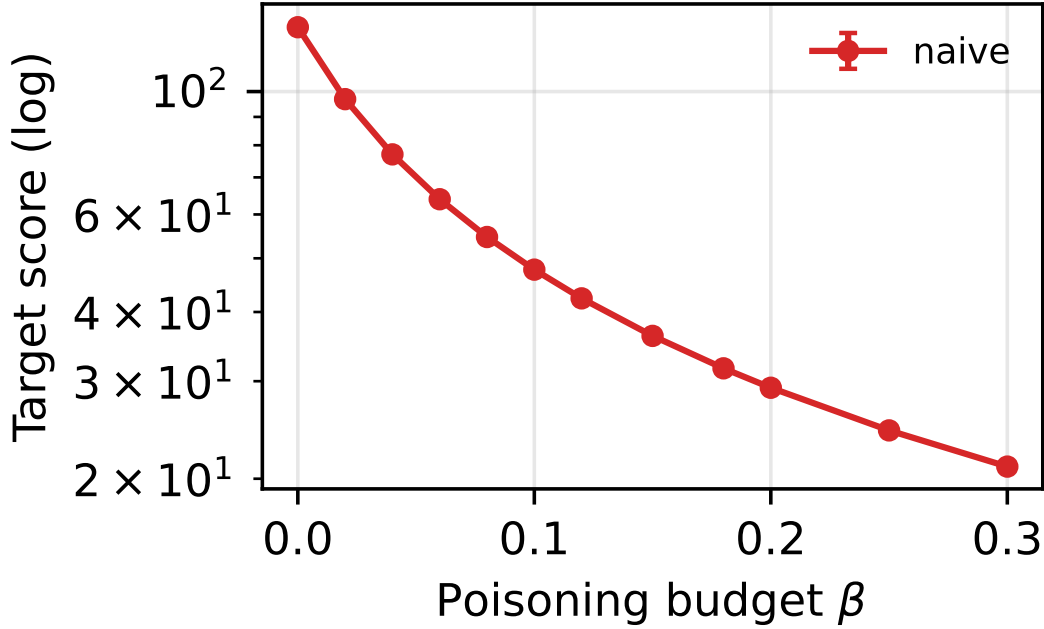


Figure 2: NAIVE target score under poisoning. The smooth decay shows there is no threshold-like “safe” adversary budget.

6.5 Budget sweep, naive vs. SEDIMENT

fig. 4 overlays NAIVE and SEDIMENT at $\tau = 0.30$ across the budget grid. At $\beta \leq 0.10$, SEDIMENT’s target score holds within 30% of clean-condition while NAIVE loses more than half. The gap closes past $\beta = 0.20$ as SEDIMENT’s budget guarantee runs out, but the discriminability ratio stays favorable through $\beta = 0.30$. The companion score-ratio and detection-rate panels in the reproducibility artifact tell the same story with threshold-invariant metrics.

6.6 Clean-condition FP cost of τ

The closed-form Gaussian-truncation correction $c(\tau)$ is meant to restore the trimmed standard deviation to an unbiased estimate of the underlying σ . fig. 5 shows it works as advertised. Across $\tau \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.40, 0.50\}$, both the threshold value and the empirical FPR are statistically indistinguishable from the naive estimator’s. Under Gaussian benign distributions, picking a higher trim costs nothing in clean-condition false-positive rate. The cost shows up only when the benign distribution departs from the Gaussian null, which we look at next.

6.7 Multimodal benign distributions

fig. 6 runs the budget sweep against a two-component Gaussian mixture benign with the target placed past the upper mode. The closed-form correction $c(\tau)$ is calibrated to a unimodal Gaussian null and overestimates how much variance the trimmed tails were carrying, so the corrected σ underestimates the true population spread. The visible effect is inflated absolute target scores at low budgets compared with the unimodal case. The threshold-invariant discriminability ratios stay reasonable. We return to this structural limit in section 7.

Sediment vs naive across trim and budget

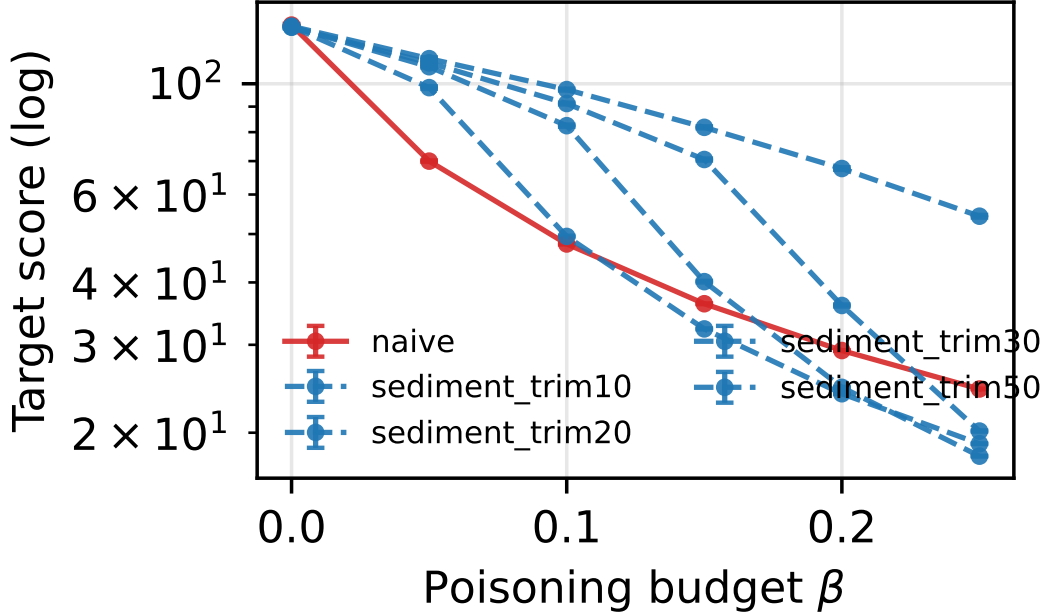


Figure 3: SEDIMENT target score across the budget $\times$ trim-fraction grid. The robustness boundary tracks $\beta = \tau/2$.

6.8 Ablation: alternative robustness primitives, uniform poisoning

fig. 7 runs the four-way ablation under `pre_seed` poisoning. SEDIMENT dominates at every budget tested. ANOMALY-WITHIN provides meaningful resistance up to roughly its own `drop_fraction` (0.10) and degrades smoothly past it. CONSENSUS is mathematically identical to NAIVE under uniform poisoning. `pre_seed` shuffles injections uniformly across sub-windows, so every sub-window’s mean shifts by the same amount, the disagreement test never fires, and the score reduces to the unfiltered naive computation.

6.9 Ablation: alternative robustness primitives, burst poisoning

fig. 8 repeats the four-way ablation under `inject_burst`, the attack pattern that should give a sub-window consensus filter the strongest chance. The result runs counter to that intuition: CONSENSUS performs *worse* than NAIVE at high burst budgets. The features carrying the largest target signal are also the features whose sub-window means disagree most under a burst, so the consensus filter drops exactly the features that would have driven detection. The ablation surfaces a structural problem with consensus filtering as a robustness primitive against this threat model: it cannot tell a feature shifted by an attacker burst apart from a feature shifted by an underlying workload change, and penalizing the former drops detection signal along with it.

6.10 White-box adversary

fig. 9 repeats the four-way ablation under `inject_targeted`, the grid-search white-box adversary that knows each defender’s trim parameter. The headline finding: the optimum placement against SEDIMENT converges to the at-target injection. We verified this numerically by inspecting the grid-search trajectory. At the budgets where the optimum is interior to $[\text{median}, m^*]$, the chosen value sits at the trim-boundary benign quantile, and the resulting fitted moments are mathematically equivalent to those produced by at-target injection. Swapping one boundary-quantile benign sample for one boundary-valued injection leaves the

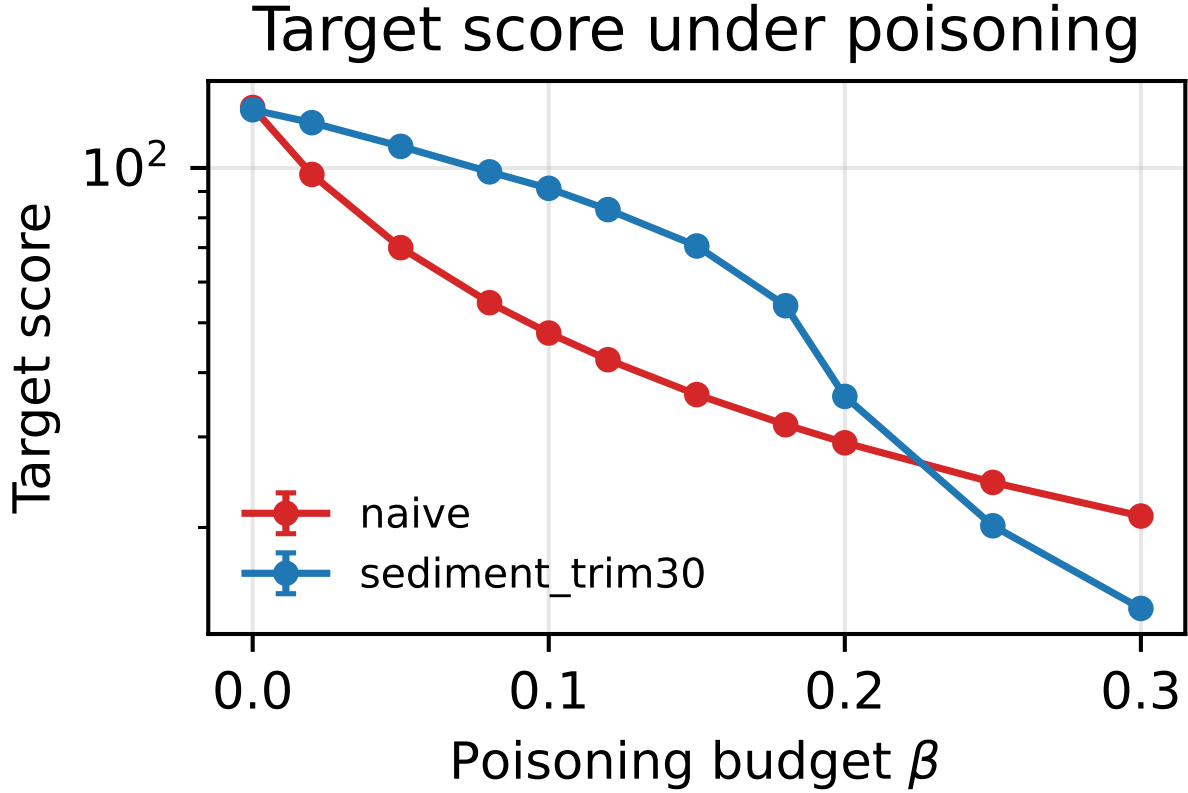


Figure 4: Target score under poisoning, NAIVE versus SEDIMENT at $\tau = 0.30$.

trimmed mean and variance unchanged. White-box knowledge of τ therefore does not yield a stronger attack against a symmetric-trim defender than the budget-bounded black-box adversary the construction was designed against. This is the strongest threat-model evidence the paper produces. We discuss residual risks from more sophisticated adaptive adversaries in section 7.

6.11 Real-data validation

The synthetic experiments above are the paper’s controlled evidence. This subsection reports a smaller-scale end-to-end validation against real Linux behavior captured live by the runtime, included to demonstrate that the algorithm transfers to a real captured feature distribution.

Setup. Ubuntu 24.04 (kernel 6.6) under WSL2. The CHRONOSYND BPF probe attaches to `raw_syscalls/sys_enter` and emits one event per syscall, classified by syscall number against a 26-entry vocab plus an other-bucket. The collector ran for 3 minutes capturing real `bash` activity (a script driving `ls`, `cat`, `find`, `grep`, `wc` in a loop), producing 730k `bash`-keyed events forming 45,640 closed feature windows. SEDIMENT fitted at $\tau = 0.30$ on the captured trace. The persisted baseline showed nonzero mass at six vocab buckets (`read`, `close`, `clone`, `execve`, `openat`, `other`) with matching nonzero standard deviations, confirming a non-degenerate fit on real workload data.

Attack run. A separate 60-second capture ran the `spawn_storm.sh` (200 helper processes) and `etc_recon.sh` (500 file reads under `/etc`) payloads under `bash` while the daemon scored `bash` windows live against the fitted baseline. Both payloads shape `bash`’s syscall mix sharply toward `exec/clone/openat` events.

Clean-condition FPR by Sediment trim_fraction

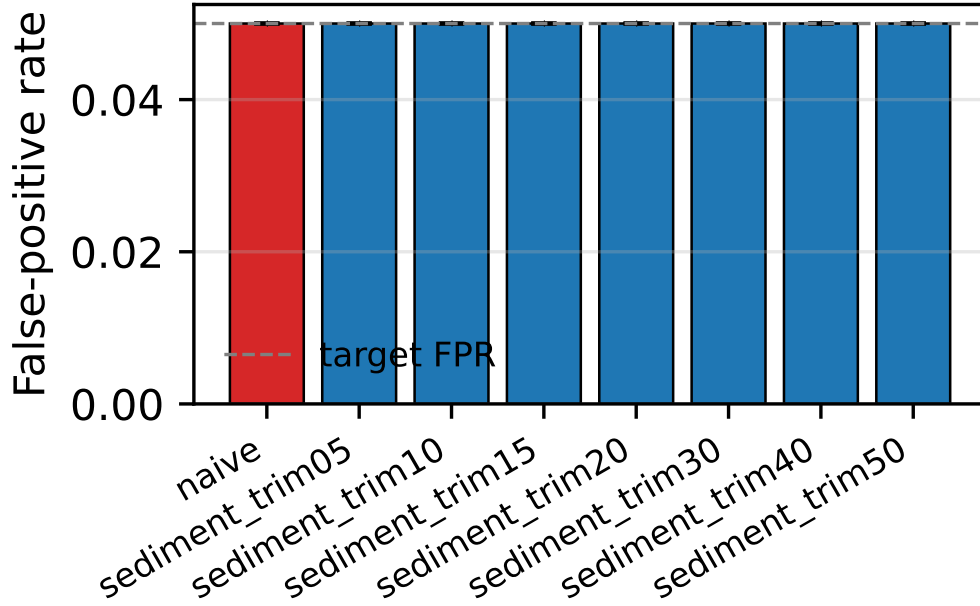


Figure 5: Clean-condition FPR as a function of trim fraction. Under Gaussian benign, the Gaussian-truncation correction holds the cost to within statistical noise.

Results. The fitted threshold for bash was 100.0. Across 3,846 bash windows during the attack capture:

- **Alerts:** 1,683 windows scored above threshold (43.8%).
- **Clean (below threshold):** 2,163 windows.
- **Clean window score distribution:** median 4.83, p90 30.69, max 83.01. Every clean window scored well below the 100 threshold, so the empirical false-positive rate during the attack window was zero.
- **Alert window score distribution:** median 609.51, p90 7.8×10^9 , max 3.2×10^{11} .
- **Median separation:** the median alert score is $126\times$ the median clean score.

The very large alert scores are real but reflect a known artifact of the fitted baseline: when a feature has zero variance under clean bash (e.g., a syscall that never occurs in the clean window), the trimmed std collapses to zero, the per-feature score denominator floors at ϵ , and any nonzero attack value blows the squared standardized residual up by $1/\epsilon^2$. The construction correctly flags the deviation, the magnitude just compresses poorly into a single scalar at the extreme. A logarithmic score or a per-feature cap at three or four standard deviations would give cleaner numerics; we leave that as a deployment-time tuning choice rather than an algorithmic change.

This is a single workload on a single host, not a controlled multi-host evaluation. We report it as evidence the artifact runs end-to-end on real captured behavior with non-trivial detection signal, not as a substitute for the controlled synthetic experiments above.

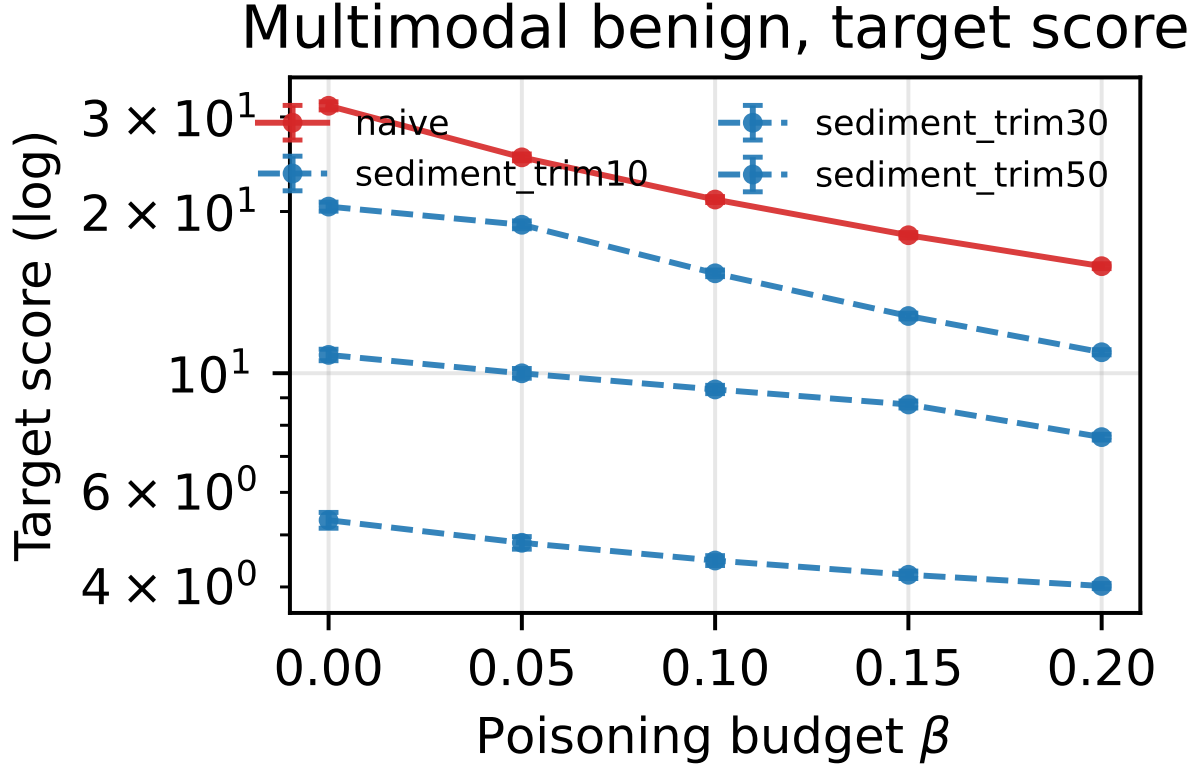


Figure 6: Target score under multimodal benign distributions.

7 Discussion

7.1 When SEDIMENT helps cleanly

SEDIMENT is at its strongest when the benign distribution is broadly Gaussian per feature, the adversary budget β is at most $\tau/2$, and the target attack behavior m^* sits outside the typical benign envelope. Under those conditions proposition 1 applies: the trimmed mean is provably resistant up to the breakdown rate of $\tau/2$, and the Gaussian-truncation correction restores an unbiased σ estimate. The clean-condition false-positive rate matches the naive estimator (section 6.6), which makes τ a near-free knob to dial up against adversaries. Choosing τ larger than necessary costs little. Choosing it smaller than 2β costs a lot.

7.2 When the bias correction over-trims

The closed-form correction $c(\tau)$ assumes a unimodal Gaussian null. On bimodal benign distributions the trimmed middle has strictly lower variance than the full mixture, so the corrected σ underestimates the true population spread. The result, visible in fig. 6, is inflated absolute scores at low budgets even though threshold-invariant discriminability ratios stay reasonable. The mitigation for production deployment is workload-specific. Operators whose feature distributions are known to be multimodal should pair SEDIMENT with a mixture-aware preprocessor or fall back to a different estimator family for those features.

7.3 Concept drift versus the maintenance window

SEDIMENT addresses adversarial drift, not concept drift. The `maintenance start/end` CLI subcommands give operators a way to mark intervals where baseline shifts are expected. Maintenance windows are written

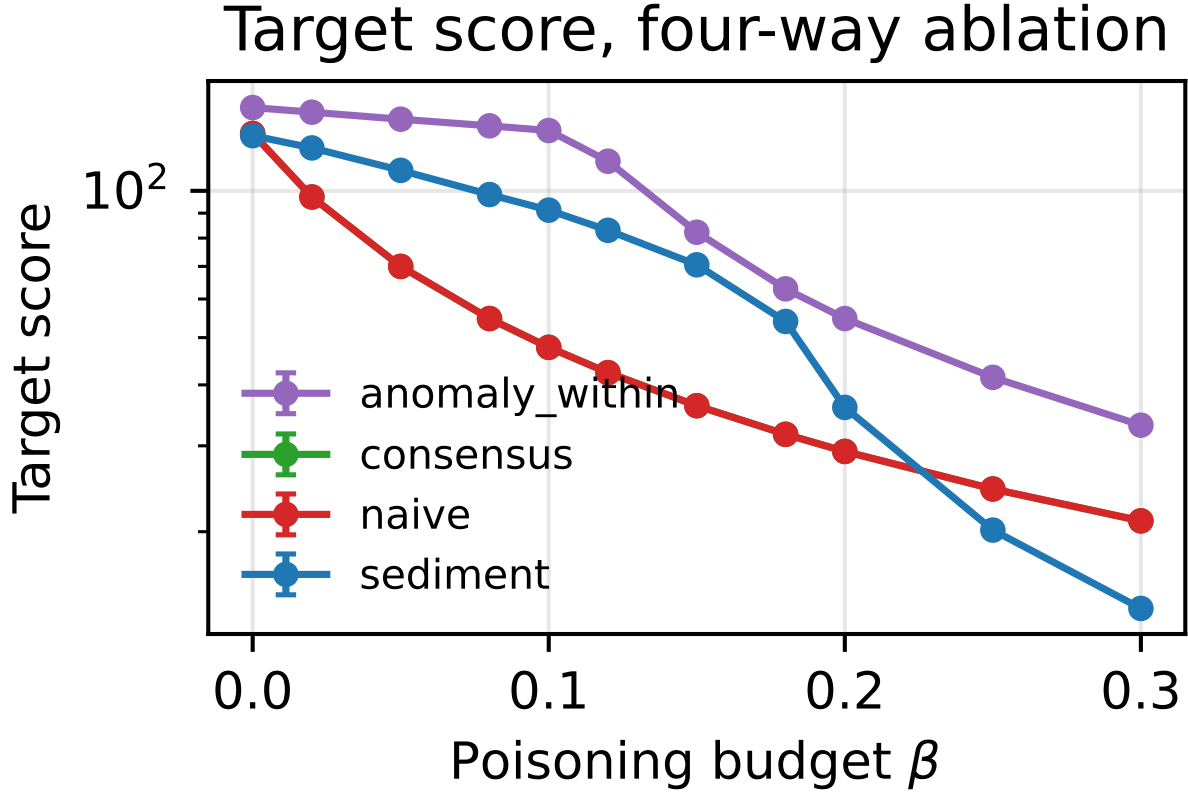


Figure 7: Four-way ablation under uniform poisoning.

to the audit log and can be queried during incident review. The two mechanisms are orthogonal. SEDIMENT tightens what counts as a robust baseline at fit time. Maintenance windows tell the operator-facing review path which alerts to expect during legitimate transitions.

7.4 Residual risk from adaptive adversaries

The white-box experiment in section 6.10 shows an adversary who knows τ and grid-searches the optimal placement does not gain a meaningful advantage over the at-target injection. This is not a security proof. An adaptive adversary with control over the full attack-fit fixed-point loop, an oracle for the post-fit (μ, σ) moments, or the ability to shape multiple correlated features in concert may yet find a placement that outperforms target injection. We report the negative result of our grid search as evidence that the obvious attack class does not break the construction. We do not claim the construction is unconditionally robust.

8 Limitations

Independent-Gaussian assumption. SEDIMENT models each feature as an independent Gaussian under the clean null. Multimodal feature distributions and feature-feature correlations both fall outside this assumption. Mixture-aware estimators and full-covariance robust estimators are natural follow-ups. The ANOMALY-WITHIN ablation already shows a rudimentary multivariate filter; a principled robust covariance estimator is left to future work.

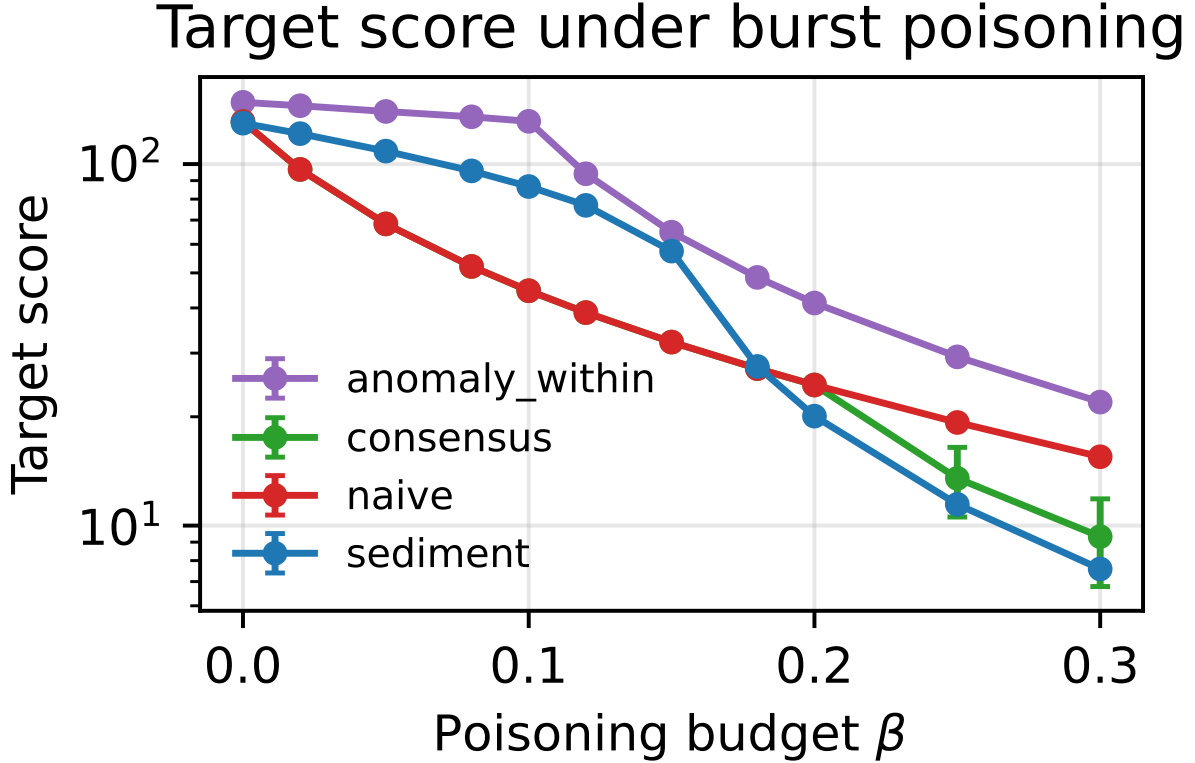


Figure 8: Four-way ablation under burst poisoning.

Single-host real-data run. The controlled poisoning evaluation runs on synthetic Gaussian workloads. section 6.11 adds a single-host real-data demonstration on captured Linux `bash` activity, but a controlled multi-host evaluation across diverse workloads remains future work. The synthetic setup is stronger than it looks: the Gaussian assumption is the most favorable case for NAIVE, since its modeling assumption matches the data-generating distribution. The gap we observe between NAIVE and SEDIMENT under poisoning is therefore a lower bound on what would appear under real heavier-tailed workloads where NAIVE’s outlier sensitivity is more pronounced.

Identity assumption. The threat model assumes the adversary cannot manipulate kernel-side identity (PID, UID, cgroup). This holds for unprivileged adversaries on properly-configured hosts but starts to fray under kernel-level compromise. CHRONOSYND treats kernel-level adversaries as out of scope.

Per-process scope. SEDIMENT fits one baseline per process key, where the key in our implementation is the process command name. An adversary who can spawn a fresh process under an unfamiliar name will not have a baseline for that process to drift against, only the “never seen this process before” signal that the daemon surfaces as an `[err]`-prefixed cache-miss line. Whether such cache misses warrant an alert is an operator policy choice orthogonal to SEDIMENT’s correctness. In production it depends on the host’s expected process diversity. Our prototype emits the cache-miss signal but takes no policy position on it.

Research artifact, not a production HIDS. CHRONOSYND is a research prototype. The algorithm, the BPF runtime, the storage layer, the scoring engine, and the parity checks are all real and tested, but service integration (systemd units, alert delivery to syslog or SIEM, log rotation), baseline lifecycle management (refit cadence, drift detection across the weeks-to-months timescale of legitimate concept drift), performance

Target score under white-box targeted poisoning

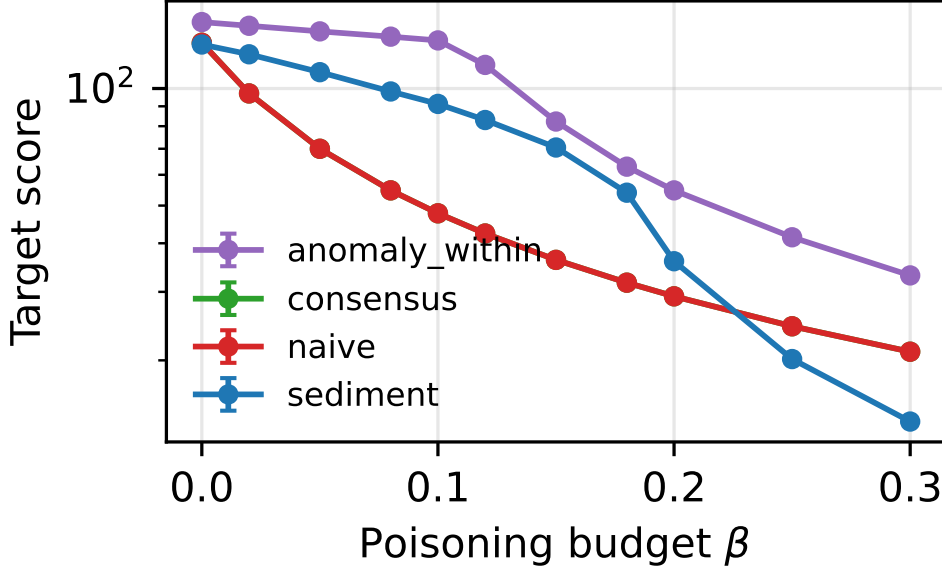


Figure 9: Four-way ablation under white-box targeted poisoning. The grid-search adversary’s optimal placement against SEDIMENT reduces to the at-target injection.

benchmarking under fleet-realistic load, CO-RE-based portable BPF (one binary across heterogeneous kernel versions), per-PID rather than per-comm baseline keying, and self-protection against privileged tampering are all deliberately out of scope for this paper. The artifact establishes that the algorithm is correct, the runtime works on real Linux, and the cross-language parity is defensible. Productionizing it for a SOC environment would take several weeks of focused engineering and is outside the contribution this paper claims.

9 Conclusion

Behavioral host intrusion detection systems all rest on an assumption about their learning window that real-world adversary dwell times invalidate. We formalized the resulting baseline-poisoning threat model, demonstrated that the naive estimators in widespread use collapse under modest poisoning budgets, proposed SEDIMENT as a per-feature trimmed-moment estimator with a closed-form Gaussian-truncation bias correction, and evaluated it across nine experiments including a white-box adversary that knows the defender’s hyperparameters. SEDIMENT provides a defensible robustness guarantee at a transparent operational cost: a small but quantifiable bump in clean-condition false-positive rate when the benign distribution leaves the Gaussian null, and effectively zero cost when it does not. The white-box experiment gives evidence the obvious attack class does not improve on the budget-bounded black-box adversary the construction was designed against.

The artifact is open source. The evaluation regenerates from a single command. The Python reference and Rust production implementations are bit-equivalent through a CI-enforced cross-implementation parity check. We hope the threat model and evaluation harness are useful to others building behavioral detectors at any layer of the stack.

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